

Step 1: Search for Windows PowerShell & click on “Run As Administrator”

Step 2:

```
C:\Users\ANANYA ROY>wsl --install
A distribution with the supplied name already exists. Use --name to chose a different name.
Error code: Wsl/InstallDistro/ERROR_ALREADY_EXISTS
```

Step 3: Restart Your PC

After the installation is complete, restart your computer if asked.

Step 4: Set Up Ubuntu

After restarting:

1. Open Ubuntu from the Start menu.
2. Wait for the setup to complete.
3. Choose a Linux username.

Example:

Enter new UNIX username: ananya

Step 5: Set a Password

Create a password and enter it again to confirm.

Example:

New password: *****

Retype new password: *****

```
ananya_roy@Ananya:~$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu resolute InRelease
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages
[492 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packa
ges [243 kB]
Fetched 873 kB in 2s (537 kB/s)
74 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Step 7: Install GCC(C Compiler)

```

ananya_roy@Ananya:~$ sudo apt install build-essential
Installing:
  build-essential

Installing dependencies:
  bzip2                libalgorithm-diff-xs-perl
  dpkg-dev              libalgorithm-merge-perl
  fakeroot              libcrypt-dev
  g++                   libdpkg-perl
  g++-15                libfakeroot
  g++-15-x86-64-linux-gnu  libfile-fcntllock-perl
  g++-x86-64-linux-gnu  libstdc++-15-dev
  libalgorithm-diff-perl  lto-disabled-list

Suggested packages:
  bzip2-doc                g++-multilib      bzip
  debian-keyring           g++-15-multilib  libstdc++-15-doc
  debian-tag2upload-keyring gcc-15-doc

Summary:
  Upgrading: 0, Installing: 17, Removing: 0, Not Upgrading: 74
  Download size: 19.1 MB
  Space needed: 72.9 MB / 1024 GB available

Continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libstdc++-15-dev amd64 15.2.0-16ubuntu1 [2920 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++-15-x86-64-linux-gnu amd64 15.2.0-16ubuntu1 [14.4 MB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++-15 amd64 15.2.0-16ubuntu1 [26.7 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++-x86-64-linux-gnu amd64 4:15.2.0-5ubuntu1 [968 B]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 g++ amd64 4:15.2.0-5ubuntu1 [1100 B]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdpkg-perl all 1.23.7ubuntu1 [289 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 bzip2 amd64 1.0.8-6build2 [34.3 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 lto-disabled-list all 79 [14.2 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/main amd64 dpkg-dev all 1.23.7ubuntu1 [1862 kB]

```

Step 8: Verify the installation

```

ananya_roy@Ananya:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

```

GITHUB REPOSITORY

ananya-964 / kernel_Codex_OSSP

Code

Issues

Pull requests

Actions

Projects

Wiki

Security and quality

Insights

Settings

Watch 0

Fork 0

0 stars

0 forks

0 watching

1 branch

0 tags

Activity

Public repository

main

Go to file

+

Code

ananya-964

Delete ForgeOS/.gitkeep

5929c63 · 3 weeks ago

2520030471_Practical	Create .gitkeep	3 weeks ago
2520030471_Skill	Create .gitkeep	3 weeks ago
2520030504_Practical	Create .gitkeep	3 weeks ago
2520030504_Skill	Create .gitkeep	3 weeks ago
ForgeOS	Delete ForgeOS/.gitkeep	3 weeks ago

Fork Process

```

#include<unistd.h>
#include<stdio.h>
int main()
{
int pid;
pid=fork();
if(pid==0)
{
printf("Child Process\n");
printf("Child process id is %d\n",getpid());
}
else if(pid>0)
{
printf("Parent Process\n");
printf("Parent process id is %d\n",getppid());
}
else if(pid<0)
{
printf("Failed process\n");
}
}
}

```

```

ananya_roy@Ananya:~$ nano skill1.c
ananya_roy@Ananya:~$ gcc skill1.c
ananya_roy@Ananya:~$ ./a.out
Parent Process
Parent process id is 316
Child Process
Child process id is 1209

```

exec() Process

```

#include<unistd.h>
int main()
{
char *args[]={"ls","-l",NULL};
execvp("ls",args);
}

```

```

ananya_roy@Ananya:~$ nano exec2.c
ananya_roy@Ananya:~$ gcc exec2.c
ananya_roy@Ananya:~$ ./a.out
total 172
-rwxr-xr-x 1 ananya_roy ananya_roy 16008 Aug 23 14:33 a.out
-rw-r--r-- 1 ananya_roy ananya_roy 1 Jul 29 06:52 a10
-rw-r-xr-x 1 ananya_roy ananya_roy 0 Jul 30 03:49 a11
-rw-r--r-- 1 ananya_roy ananya_roy 54 Jul 29 06:55 a2.c
-rw-r--r-- 1 ananya_roy ananya_roy 53 Jul 30 04:19 a7
-rw-r--r-- 1 ananya_roy ananya_roy 70 Jul 29 06:03 a8
-rw-r--r-- 1 ananya_roy ananya_roy 17 Jul 29 06:08 a9
-rwxrwxrwx 1 ananya_roy ananya_roy 0 Jul 29 08:26 abc
-rw-r--r-- 1 ananya_roy ananya_roy 20 Aug 1 05:41 b1
-rw-r-xr-x 1 ananya_roy ananya_roy 10 Aug 1 05:56 b2
d-wxr-xr-x 2 ananya_roy ananya_roy 4096 Jul 29 08:15 dirz
-rw-r--r-- 1 ananya_roy ananya_roy 114 Aug 12 04:50 ex1
-rw-r--r-- 1 ananya_roy ananya_roy 85 Aug 23 14:32 exec2.c
-rw-r--r-- 1 ananya_roy ananya_roy 68 Aug 12 07:57 execex1.c
-rw-r--r-- 1 ananya_roy ananya_roy 63 Aug 12 06:50 execex2.c
-rw-r--r-- 1 ananya_roy ananya_roy 139 Aug 12 07:10 execvex3.c
-rwxr-xr-x 1 ananya_roy ananya_roy 15992 Jul 29 06:49 f1
-rw-r--r-- 1 ananya_roy ananya_roy 124 Jul 29 07:56 f1.c
drwxr-xr-x 2 ananya_roy ananya_roy 4096 Jul 29 16:06 linux_task1
drwxr-xr-x 3 ananya_roy ananya_roy 4096 Jul 29 16:20 linux_task2
drwxr-xr-x 2 ananya_roy ananya_roy 4096 Jul 29 16:23 linux_task3
drwxr-xr-x 2 ananya_roy ananya_roy 4096 Jul 29 16:28 linux_task4
-rw-r--r-- 1 ananya_roy ananya_roy 291 Jul 30 04:19 open1.c
drwxr-xr-x 5 ananya_roy ananya_roy 4096 Aug 23 13:45 ossp
drwxr-xr-x 2 ananya_roy ananya_roy 4096 Jul 30 06:27 permission_lab
-rw-r--r-- 1 ananya_roy ananya_roy 247 Aug 12 08:17 pipe1.c
-rw-r--r-- 1 ananya_roy ananya_roy 249 Aug 12 08:36 pipe2.c
-rw-r--r-- 1 ananya_roy ananya_roy 251 Aug 12 08:39 pipe3.c
-rwxr-xr-x 1 ananya_roy ananya_roy 16088 Aug 12 05:27 pipeex1
-rw-r--r-- 1 ananya_roy ananya_roy 219 Aug 12 05:24 pipeex1.c
-rw-r--r-- 1 ananya_roy ananya_roy 275 Aug 12 06:09 pipeex2.c

```

```
#include<stdio.h>
#include<string.h>
int main()
{
char input[100];
while(1)
{
printf("myshell> ");
scanf("%s", input);
if(strcmp(input, "exit")==0)
{
printf("Goodbye!\n");
break;
}
printf("You entered: %s\n",input);
}
}
```

```
ananya_roy@Ananya:~$ nano exec2.c
ananya_roy@Ananya:~$ nano skill2.c
ananya_roy@Ananya:~$ gcc skill2.c
ananya_roy@Ananya:~$ ./a.out
myshell> hii
You entered: hii
myshell> cse
You entered: cse
myshell> earth
You entered: earth
myshell> exit
Goodbye!
```

